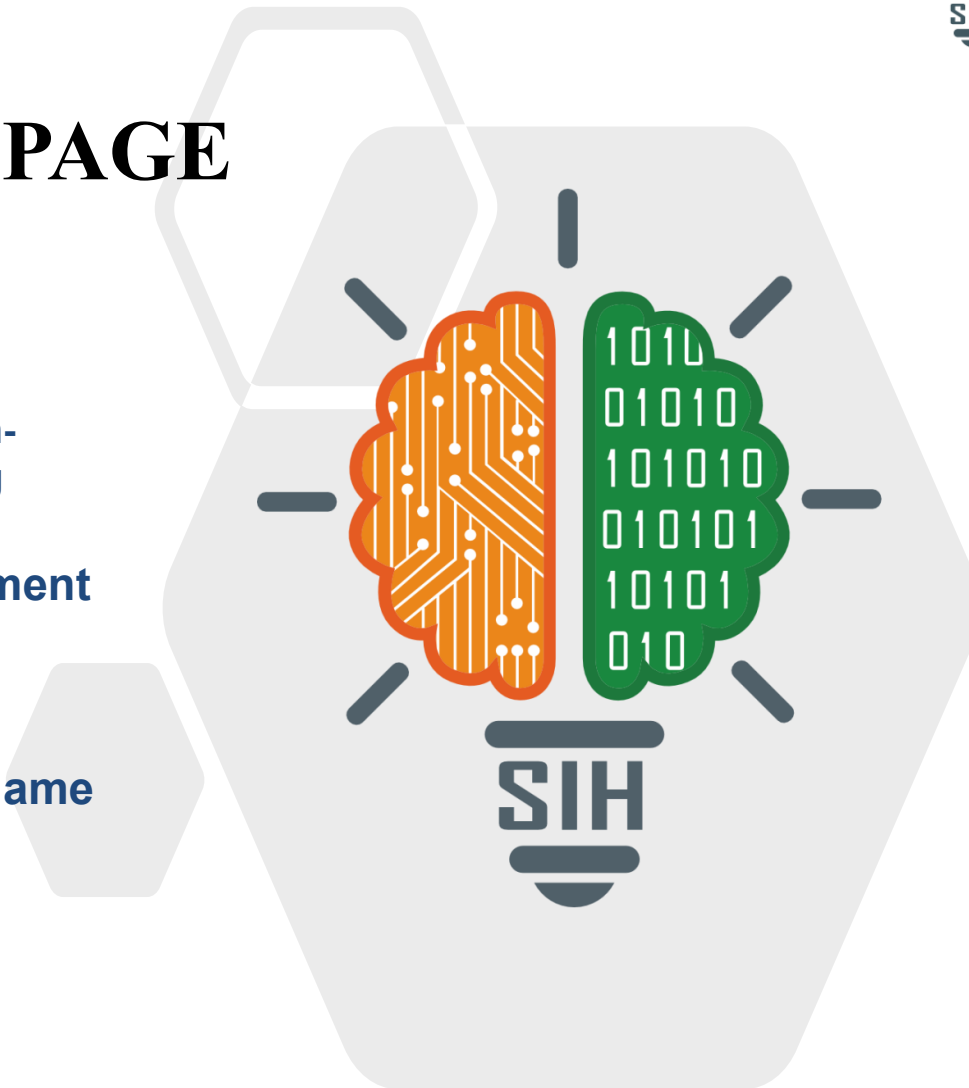


SMART INDIA HACKATHON 2026



TITLE PAGE

- Problem Statement ID – **SIH26021**
- Problem Statement Title- **Honey Chain: A block chain-based system for honey traceability and smart beekeeping management.**
- Theme- **Agriculture, FoodTech & Rural Development**
- PS Category- **Software**
- Team ID- **<Team ID>**
- Team Name (Registered on portal)- **Your Team Name**



HONEY CHAIN – Blockchain-verified honey, hive to shelf

❖ Proposed Solution (Describe your Idea/Solution/Prototype)

Detailed explanation of the proposed solution

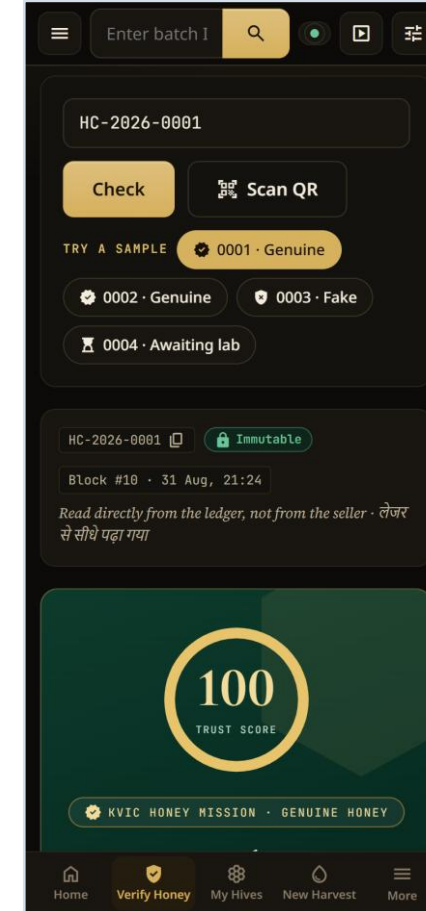
- Every hive gets a digital ID; every harvest becomes a QR-coded batch
- Labs seal purity results (incl. NMR + rice-syrup marker) on the ledger
- Buyer scans the QR → origin, lab report, journey, 0-100 trust score
- IoT + AI monitor hive health and forecast yield; data anchored on-chain

How it addresses the problem

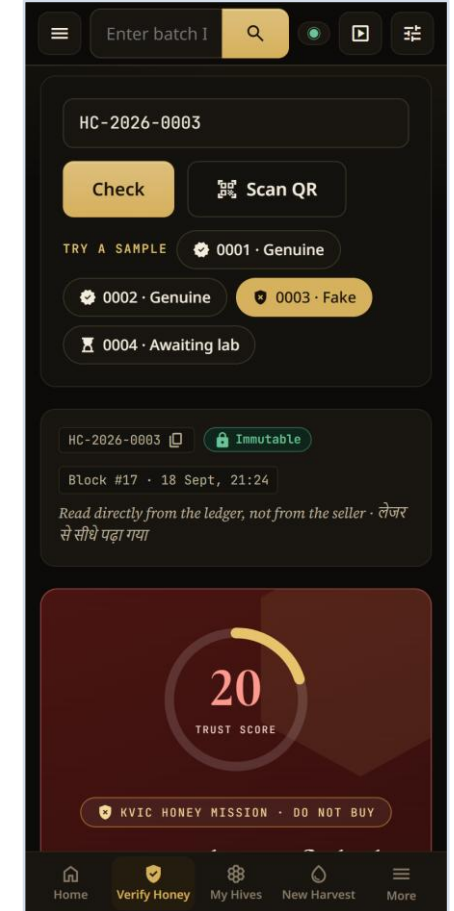
- No lab certificate on the ledger = no trust score, so fakes cannot pass
- Yield is cross-checked with hive data: "95 kg from 1 hive" gets flagged
- Provable quality earns beekeepers premium prices and direct buyers

Innovation and uniqueness of the solution

- Lab-agnostic trust layer: works with NMR, SMR, C4 and HMF results
- Sensor-anchored yield check catches blending that chemistry can miss
- Offline-first bilingual PWA + zero-fee permissioned chain for villages



Genuine batch
Trust score 100



Fake batch
Trust score 20 - do not buy

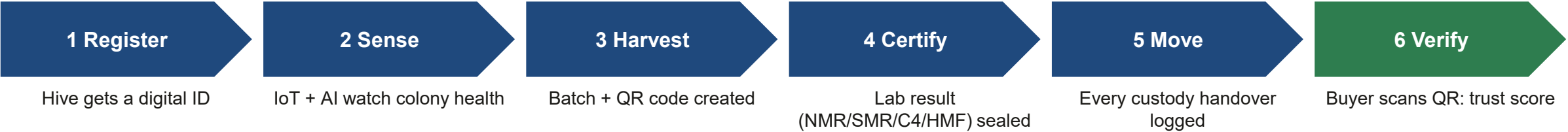
❖ Technologies to be used (e.g. programming languages, frameworks, hardware)

Frontend	React 18 · Vite · offline-first installable PWA · QR camera scan · Web Crypto
Ledger	SHA-256 hash-chain · Proof-of-Authority validators · Merkle roots · Solidity → Hyperledger Besu
IoT	ESP32 · load cell · DS18B20 · DHT22 · MEMS mic · CO ₂ · LoRa · solar
AI	Explainable feature model (prototype) → CNN on hive audio via TensorFlow Lite Micro
DevOps	GitHub Actions CI · GitHub Pages / Vercel / Docker (nginx) · service-worker caching



Working prototype: live hive dashboard - AI flags "Pre-swarm 98%" with advice and yield forecast

❖ Methodology and process for implementation (Flow Charts/Images/ working prototype)



Only hashes go on-chain; raw sensor data stays local and is anchored as a Merkle root, so a 2G link is enough.

Analysis of the feasibility of the idea

- Working prototype already live: ledger, QR verify, lab portal, hive AI
- Estimated hardware ₹1.5-2.5k per hive; one LoRa gateway serves 20-40 hives*
- Permissioned PoA: no mining, no gas; validators are KVIC, FSSAI and labs
- Fits existing NBHM / KVIC clusters and NABL-accredited labs

Potential challenges and risks

- Garbage-in: a dishonest lab or beekeeper can enter false data
- Engineered syrups can pass basic tests (CSE 2020: NMR failures)
- Patchy connectivity and low digital literacy
- QR cloning and jar refilling

Strategies for overcoming these challenges

- Accredited-lab keys, random audits, IoT weight vs claimed-yield check
- Anchor NMR + rice-syrup marker results; design stays lab-agnostic
- Offline-first bilingual PWA, QR-first flows, LoRa gateways
- Per-jar serials and single-use claim codes (roadmap)

Validation and rollout plan

Validate first: 10-minute interviews with 5 beekeepers, 2 NABL labs and 1 KVIC field officer (script prepared), then run the pilot.

Pilot · 0-3 months

2 clusters · 50 beekeepers · 100 hives · 1 lab

State · 4-9 months

10 clusters · 3 states · QR at Khadi outlets

Regional · 10-18 months

All Honey Mission states · FSSAI dashboard

National · Year 2+

Open API · export certificates · trained AI

*Planning estimates, to be validated during the pilot.

❖ Potential impact on the target audience

Beekeepers

- Proof of quality → premium price, direct buyers
- Early disease alerts cut colony loss

Consumers

- Scan-to-verify in seconds
- Fake batches flagged before purchase

KVIC & regulators

- Cluster dashboards: output, adulteration flags
- Targeted extension support

Processors & retailers

- Trusted supply chain, lower recall risk
- Verified, premium-ready listings

❖ Benefits of the solution (social, economic, environmental, etc.)

Social

- Trust in rural produce
- Livelihood security incl. SHGs
- Bilingual, low-literacy UX

Economic

- Higher realisation per kg
- Fewer fraud losses for buyers
- Export-ready traceability

Environmental

- Healthier colonies → pollination
- Targeted treatment, less chemical
- Low-energy ledger (no mining)

WHY NOW - from public sources

17 of 22

honey samples tested were adulterated with sugar syrup (77%)

CSE investigation, Dec 2020

1.4 lakh MT

natural honey produced in India in 2024

PIB - National Beekeeping & Honey Mission

₹500 crore

NBHM outlay, FY 2020-21 to 2025-26 - production is growing, trust must keep pace

PIB

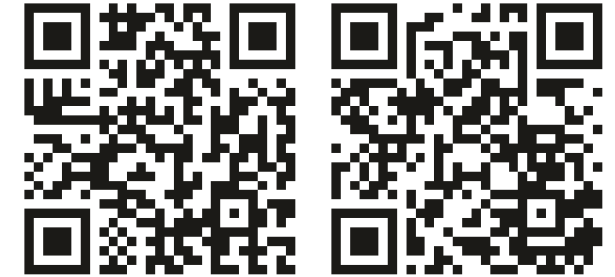
❖ Details / Links of the reference and research work

1. CSE, "Laboratory results of honey testing" (Dec 2020) - NMR tests on branded honey
cdn.downtoearth.org.in/report_laboratory_results_of_honey_testing_20201201.pdf
2. FSSAI press release reviewing the CSE honey-sample issue (3 Dec 2020)
fssai.gov.in/upload/press_release/2020/12/5fc8eb6d70965Press_Release_Honey_Sample_03_12_2020.pdf
3. PIB, "National Beekeeping & Honey Mission" - budget, production and export data
pib.gov.in/PressReleasePage.aspx?PRID=2185400
4. National Bee Board - NBHM portal
nbb.gov.in
5. The Tribune, "Honeytrap" - syrups designed to bypass purity tests
tribuneindia.com/news/nation/honeytrap-cse-investigations-claim-nefarious-adulteration-business-of-honey-designed-to-bypass-purity-tests-178930
6. Business Standard - FSSAI on the SMR (rice-syrup marker) test
business-standard.com/article/companies/fssai-questions-cse-for-not-conducting-smr-test-on-honey-brands-120120301199_1.html

Research takeaways applied in the design

- Engineered syrups can pass C4/HMF but fail NMR → the ledger is lab-agnostic and also anchors NMR + SMR results
- Domestic honey testing lacks a single verifiable trail → tamper-evident batch records and a public trust score
- Lab limits in the prototype are indicative (FSSAI/BIS-style); final limits come from the notified lab

TRY THE PROTOTYPE



Live demo

suyash2527.github.io/HoneyChain

Source code

github.com/Suyash2527/HoneyChain



Scan to verify a genuine sample batch

React · PWA · SHA-256 PoA ledger · IoT + AI simulation. MIT
licensed, open source.